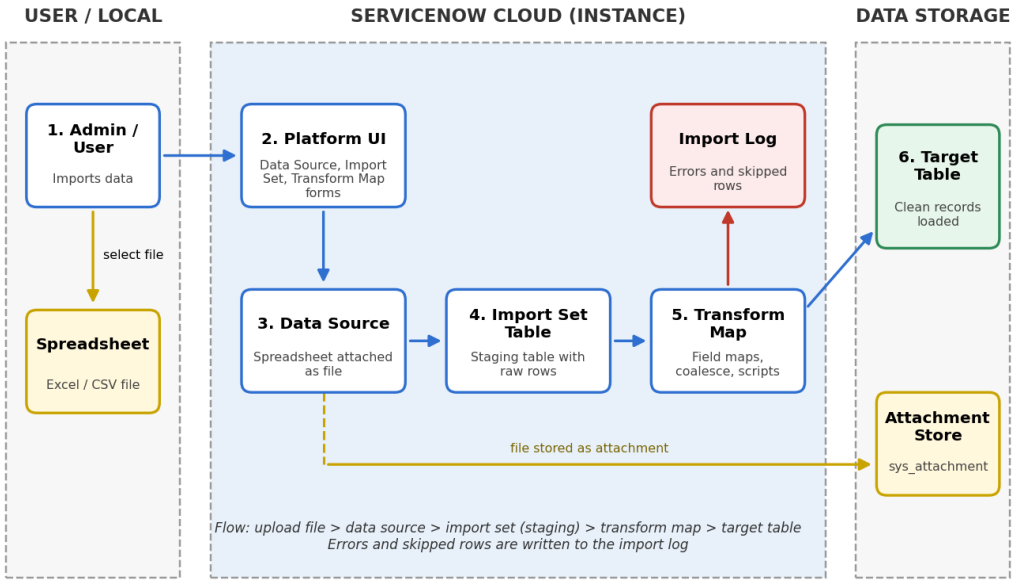
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	6ab221d587602313e82c60c9
Project Name	Import data using transform maps (spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table 1 & table 2

Example: Import data using transform maps (spreadsheet)



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	ServiceNow platform UI where the admin creates the Data Source, Import Set and Transform Map and starts the import	ServiceNow Next Experience UI (HTML, CSS, JavaScript)
2.	Application Logic-1	Data Source setup: the spreadsheet (Excel/CSV) file is attached and selected as the source of data	ServiceNow Data Source (File type, Excel/CSV)
3.	Application Logic-2	Loading the spreadsheet rows into a staging (Import Set) table	ServiceNow Import Set, Import Set Table
4.	Application Logic-3	Transform map with field maps and Coalesce to move staging data into the target table without duplicates	ServiceNow Transform Map, Field Maps, Coalesce
5.	Database	Target table that stores the final transformed records, plus staging and log tables	ServiceNow relational database (Now Platform tables)
6.	Cloud Database	Database service hosted on the cloud with the ServiceNow instance	ServiceNow Cloud (Now Platform)
7.	File Storage	Storage of the uploaded spreadsheet file used by the data source	ServiceNow Attachments (sys_attachment)
8.	External API-1	Optional: REST API to push spreadsheet data into an Import Set table from outside	ServiceNow Import Set REST API (optional)
9.	External API-2	Optional: MID Server for data that is not directly reachable from the cloud instance	ServiceNow MID Server (optional)
10.	Machine Learning Model	Not used in this project	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Cloud server configuration: ServiceNow developer instance. Local server configuration: not used	ServiceNow Personal Developer Instance (PDI), cloud hosted

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open file formats and open scripting used: spreadsheet files (.xlsx/.csv) and JavaScript in transform scripts	JavaScript, CSV / XLSX formats

S.No	Characteristics	Description	Technology
2.	Security Implementations	Role-based access (only admin / import_admin roles can import), ACLs on tables, secure HTTPS access to the instance	Roles, ACLs, HTTPS / TLS
3.	Scalable Architecture	Import sets are staged and transformed in batches on a cloud multi-tier platform, so large files can be handled	ServiceNow multi-tier cloud architecture
4.	Availability	Cloud instance hosted with redundant data centers and failover, so import can be run whenever needed	ServiceNow cloud high availability
5.	Performance	Bulk load through staging table, indexed Coalesce field for fast matching, and scheduled or background transform for big files	Import Set batching, indexes, Scheduled Import